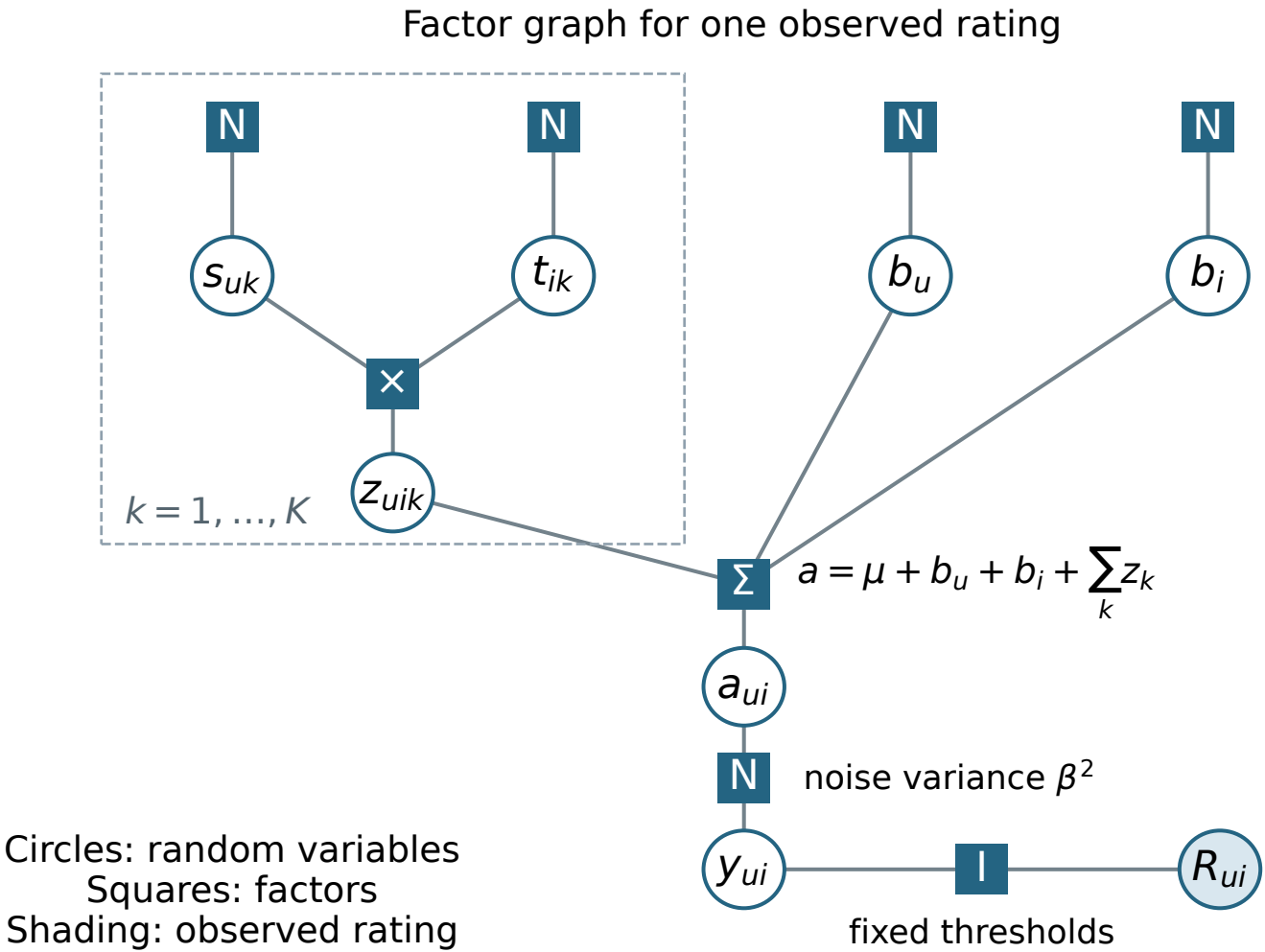
Factor graph for one observed rating
N
N
N
N
s_{uk}
t_{ik}
b_u
b_i
×
z_{uik}
$k = 1, \ldots, K$
Σ
$a = \mu + b_u + b_i + \sum_k z_k$
a_{ui}
N
noise variance β^2
y_{ui}
I
R_{ui}
fixed thresholds
Circles: random variables
Squares: factors
Shading: observed rating